Here’s a complete research paper draft you can directly use for your 2nd project.

System Capacity & Care Load Analytics for Unaccompanied Children

Abstract

The Unaccompanied Alien Children (UAC) Program is a federally mandated initiative responsible for providing temporary care and support services to children transferred from U.S. Customs and Border Protection (CBP) to the Department of Health and Human Services (HHS). Due to fluctuating border activity and humanitarian conditions, the program experiences varying levels of care demand and operational stress. This project develops a healthcare analytics framework to monitor system capacity, evaluate care load dynamics, and identify periods of strain and relief within the UAC care pipeline. Using time-series analysis and healthcare capacity metrics, the study analyzes trends in CBP custody, HHS care load, transfers, and discharges. The results provide actionable insights for policymakers and healthcare planners to improve operational readiness, resource allocation, and humanitarian response effectiveness.

---

1. Introduction

The UAC Program functions as a dynamic healthcare and welfare pipeline for children entering federal custody. Children apprehended by CBP are transferred to HHS facilities where they receive medical screening, sheltering, mental health support, and reunification services before being discharged to approved sponsors.

Managing this system requires continuous monitoring of capacity utilization and operational load. Sudden increases in child intake can strain shelter infrastructure, healthcare staffing, and case management resources. Therefore, healthcare analytics and data-driven monitoring systems are essential for maintaining sustainable service delivery.

This project focuses on analyzing system capacity and care load trends using operational data collected between 2023 and 2025. By converting raw operational counts into meaningful healthcare metrics, the study enables improved situational awareness and proactive decision-making.

---

2. Problem Statement

Although the UAC Program collects large amounts of operational data daily, there is currently no centralized analytical framework for evaluating:

- Total system care load
- Balance between intake and discharge

- Capacity strain and relief periods
- Long-term sustainability of care delivery

Without structured analytics, policymakers and operational teams face challenges in identifying periods of overcrowding, staffing stress, and resource shortages.

---

3. Objectives

Primary Objectives

- Quantify daily and cumulative care load across CBP and HHS
- Identify periods of capacity strain and operational relief
- Analyze balance between intake, transfers, and discharges

Secondary Objectives

- Support healthcare staffing and shelter planning
- Improve situational awareness for policymakers
- Enable data-driven humanitarian response evaluation

---

4. Dataset Description

The dataset consists of daily operational records related to the UAC Program.

Column Name	Description
Date	Reporting date
Children apprehended and placed in CBP custody	Daily intake volume
Children in CBP custody	Active CBP care load
Children transferred out of CBP custody	Flow into HHS system
Children in HHS Care	Active HHS care load
Children discharged from HHS Care	Successful sponsor placements

The dataset covers daily observations from 2023 to 2025.

---

5. Methodology

5.1 Data Ingestion and Structuring

The dataset was loaded into Python using the Pandas library. The Date column was converted into datetime format to support time-series analysis. Data was sorted chronologically and indexed by date.

5.2 Data Quality Validation

Data validation checks were performed to ensure consistency and transparency.

The following checks were implemented:

- Identification of missing values
- Duplicate date detection
- Logical constraint validation:
  - Transfers ≤ CBP custody
  - Discharges ≤ HHS care

Any anomalies identified during preprocessing were flagged for further inspection.

5.3 Derived Healthcare Capacity Metrics

Several analytical metrics were calculated to evaluate system performance.

Total System Load

The total system load was computed as:

CBP Custody + HHS Care

This metric represents the combined operational burden across both agencies.

Net Daily Intake

The net intake was calculated as:

Transfers into HHS − Discharges from HHS

Positive values indicate growing system pressure.

Care Load Growth Rate

Day-over-day percentage changes were calculated to measure fluctuations in total system load.

Rolling Averages

7-day and 14-day rolling averages were computed to identify sustained trends and smooth short-term variability.

---

6. Trend and Temporal Analysis

Time-series analysis revealed clear fluctuations in care load across the observation period. Several sustained high-load periods were observed, indicating increased operational stress on shelter systems and healthcare services.

Rolling average analysis showed that system pressure often remained elevated for multiple consecutive days, highlighting the importance of proactive capacity management.

Comparisons between early and later time periods demonstrated variations in intake patterns and discharge efficiency over time.

---

7. Results and Findings

The analysis produced several important findings.

7.1 Total System Load

The total system load exhibited substantial variability throughout the reporting period. Peaks in care load corresponded to periods of increased child apprehensions and transfers into HHS care.

7.2 CBP vs HHS Care Comparison

The comparison between CBP custody and HHS care load showed that HHS facilities consistently maintained higher long-term care volumes, reflecting extended shelter and placement operations.

7.3 Net Intake Trends

Positive net intake values were observed during high-demand periods, indicating that children were entering HHS care faster than they were being discharged to sponsors.

Sustained positive net intake contributed to backlog accumulation and increased operational stress.

7.4 Rolling Average Analysis

Rolling averages effectively identified prolonged periods of system strain. The 14-day rolling average provided a clearer view of sustained pressure trends compared to daily fluctuations.

---

8. Discussion

The findings demonstrate the importance of healthcare analytics in monitoring large-scale humanitarian care systems. Continuous tracking of operational metrics enables decision-makers to anticipate strain periods and allocate resources proactively.

The analysis also highlights the value of trend monitoring for identifying inefficiencies in discharge processes and shelter capacity utilization.

Implementing real-time analytical dashboards could further improve transparency and operational responsiveness.

---

9. Recommendations

Based on the findings, the following recommendations are proposed:

- Develop centralized operational dashboards for real-time monitoring
- Increase staffing flexibility during surge periods
- Expand temporary shelter capacity during sustained high-load windows
- Improve sponsor placement workflows to reduce backlog accumulation
- Implement predictive analytics for early-warning capacity alerts

---

10. Conclusion

This project developed a healthcare analytics framework for evaluating system capacity and care load dynamics within the UAC Program. By transforming raw operational data into meaningful analytical insights, the study provided a structured understanding of care system performance, operational stress, and capacity sustainability.

The findings support the use of data-driven decision-making for improving humanitarian care delivery, healthcare planning, and operational readiness within federally mandated child welfare systems.

References

1. U.S. Department of Health and Human Services (HHS)
2. U.S. Customs and Border Protection (CBP)
3. Healthcare Capacity Analytics Literature
4. Time-Series and Operational Analytics Research